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Spatiotemporal patterns of macroalgal diversity across an environmentally heterogeneous oceanic archipelago

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Abstract:	Anthropogenic climate change is altering oceanographic conditions worldwide, with particularly strong implications for coastal ecosystems and habitat-forming macroalgae. Detecting long-term patterns in macroalgal diversity under environmental variability remains challenging, especially in regions characterized by strong spatial heterogeneity. Here, we integrate nearly three decades of macroalgal occurrence records with satellite-derived environmental data to examine spatiotemporal patterns of macroalgal diversity across the Canary Islands (NE Atlantic). Using historical biodiversity records spanning 1993-2023, we quantified alpha, beta, and gamma diversity at island and regional scales, and subsequently related these patterns to physical and biogeochemical variables derived from the Copernicus Marine Environment Monitoring Service. Spatial patterns were evaluated along the pronounced east-west environmental gradient of the archipelago, while temporal trends were assessed across seasonal, interannual, and decadal scales. Relationships between macroalgal diversity and environmental variability were explored using generalized additive models and multivariate analyses. Macroalgal diversity exhibited strong spatial structuring across islands. Eastern islands, influenced by cooler and nutrient-enriched waters, and western islands, characterized by warmer and more oligotrophic conditions, showed contrasting diversity patterns and community compositions. Beta diversity analyses revealed clear compositional clustering among islands, consistent with regional environmental differentiation. Temporally, species richness increased during the 2000s but declined in subsequent decades, particularly in western islands, against a background of coherent inter-decadal changes in oceanographic conditions. Statistical models indicated that macroalgal diversity metrics were associated with multiple environmental variables through non-linear relationships, and model performance improved when analyses were stratified by region rather than conducted at the archipelago-wide scale. These patterns are compatible with a gradual reorganization of macroalgal communities under long-term environmental variability, although causal mechanisms cannot be directly inferred from the observational data. By combining long-term biological records with satellite-derived environmental information, this study provides a regional-scale baseline of macroalgal diversity dynamics in an environmentally heterogeneous archipelago. The results highlight the

	importance of spatial context and scale when assessing biodiversity responses to contemporary climate variability and offer a framework for future macro-ecological analyses in coastal systems.
Opposed Reviewers:	



Las Palmas de Gran Canaria

3rd May 2026

Dear Editor,

We are pleased to submit the manuscript entitled “***Spatiotemporal patterns of macroalgal diversity across an environmentally heterogeneous oceanic archipelago***” for consideration for publication in *Estuarine, Coastal and Shelf Science*.

In this study, we integrate nearly three decades of macroalgal occurrence records with satellite-derived physical and biogeochemical data to assess spatial and temporal patterns of macroalgal diversity across the Canary Islands. By combining long-term biodiversity data with reanalysis products from the Copernicus Marine Environment Monitoring Service, we examine alpha, beta, and gamma diversity dynamics across islands and regions, and relate these patterns to environmental variability across seasonal to decadal scales.

Our results reveal pronounced spatial structuring of macroalgal communities along the well-documented east west environmental gradient of the archipelago, as well as coherent long-term changes in diversity that coincide with sustained variability in oceanographic conditions. Using generalized additive models and multivariate analyses, we show that diversity–environment relationships are non-linear and strongly scale dependent, with improved explanatory power when analyses are stratified by environmentally distinct regions. While the observational nature of the data precludes causal inference, the patterns identified are compatible with a gradual reorganization of macroalgal communities under long-term environmental variability.

We believe that this manuscript is well suited to *Estuarine, Coastal and Marine Science* because it addresses key topics within the journal’s scope, including climate-related environmental change, long-term ecological variability, and the integration of biological and environmental datasets to understand coastal ecosystem dynamics. The study provides a regional-scale baseline that is relevant for biodiversity assessment, macro-ecological research, and adaptive management in environmentally heterogeneous marine systems.

This manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have approved the submitted version and agree with its submission to *Estuarine, Coastal and Marine Science*. Any necessary permits and ethical standards related to the data used have been respected.

Thank you for considering our manuscript. We look forward to your evaluation.

Sincerely,

Prof Rodrigo Riera (PhD)

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- Integrated three decades of macroalgal records with satellite-derived environmental data across the Canary Islands.
- Macroalgal alpha, beta, and gamma diversity structured along a pronounced east–west environmental gradient.
- Eastern and western islands showed contrasting diversity patterns under differing oceanographic conditions.
- Species richness increased during the 2000s and declined in recent decades, especially in western islands.
- Diversity metrics showed non-linear associations with multiple physical and biogeochemical variables.
- Regionally stratified models outperformed archipelago-wide models in explaining diversity–environment relationships.
- Results provide a baseline for assessing macroalgal diversity responses to long-term environmental variability.

Spatiotemporal patterns of macroalgal diversity across an environmentally heterogeneous oceanic archipelago

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ABSTRACT

Anthropogenic climate change is altering oceanographic conditions worldwide, with particularly strong implications for coastal ecosystems and habitat-forming macroalgae. Detecting long-term patterns in macroalgal diversity under environmental variability remains challenging, especially in regions characterized by strong spatial heterogeneity. Here, we integrate nearly three decades of macroalgal occurrence records with satellite-derived environmental data to examine spatiotemporal patterns of macroalgal diversity across the Canary Islands (NE Atlantic). Using historical biodiversity records spanning 1993-2023, we quantified alpha, beta, and gamma diversity at island and regional scales, and subsequently related these patterns to physical and biogeochemical variables derived from the Copernicus Marine Environment Monitoring Service. Spatial patterns were evaluated along the pronounced east-west environmental gradient of the archipelago, while temporal trends were assessed across seasonal, interannual, and decadal scales. Relationships between macroalgal diversity and environmental variability were explored using generalized additive models and multivariate analyses. Macroalgal diversity exhibited strong spatial structuring across islands. Eastern islands, influenced by cooler and nutrient-enriched waters, and western islands, characterized by warmer and more oligotrophic conditions, showed contrasting diversity patterns and community compositions. Beta diversity analyses revealed clear compositional clustering among islands, consistent with regional environmental differentiation. Temporally, species richness increased during the 2000s but declined in subsequent decades, particularly in western islands, against a background of coherent inter-decadal changes in oceanographic conditions. Statistical models indicated that macroalgal diversity metrics were associated with multiple environmental variables through non-linear relationships, and model performance improved when analyses were stratified by region rather than conducted at the archipelago-wide scale. These patterns are compatible with a gradual reorganization of macroalgal communities under long-term environmental variability, although causal mechanisms cannot be directly inferred from the observational data. By combining long-term biological records with satellite-derived environmental information, this study provides a regional-scale baseline of macroalgal diversity dynamics in an environmentally heterogeneous archipelago. The results highlight the importance of spatial context and scale when assessing biodiversity responses to contemporary climate variability and offer a framework for future macro-ecological analyses in coastal systems.

Introduction

Anthropogenic climate change is reshaping marine environments through sustained thermal shifts in sea surface temperature, altered circulation patterns, and changes in water column stratification (1; 2; 3; 4). These processes are particularly critical in coastal systems, where benthic communities experience rapid exposure to thermal stress, nutrient variability, and hydrodynamic forcing (5). As a result, many marine species are undergoing range shifts, local population declines, or compositional reorganization when environmental thresholds are exceeded (6; 7; 8).

Habitat-forming macroalgae are among the most sensitive and ecologically consequential components of coastal ecosystems under climate change (5; 9). These taxa play a central role in structuring benthic habitats, sustaining trophic interactions, and supporting ecosystem services such as fisheries production and coastal protection (10; 11). Documented responses of macroalgal communities to environmental change include distributional shifts, local extinctions, and changes in dominance patterns, often accompanied by increases in opportunistic or warm-affinity species (1; 12; 13). Understanding how macroalgal diversity responds to long-term environmental variability is therefore essential for anticipating future ecosystem trajectories under ongoing ocean warming.

Long-term ecological records provide a unique opportunity to assess such responses across decadal timescales. Herbarium collections and biodiversity databases, when combined with remotely sensed environmental data, enable the detection of

broad-scale temporal trends and spatial gradients that are otherwise difficult to capture through short-term field studies (14; 15). This integrative approach has proven particularly valuable for identifying macro-ecological patterns associated with climate variability, allowing researchers to distinguish persistent signals from natural interannual fluctuations (16).

Among statistical tools suited to this objective, generalized additive models (GAMs) offer a flexible framework for characterizing non-linear relationships between biological responses and multiple environmental drivers (17; 18). GAMs have been widely applied in marine ecology to explore species-environment associations across heterogeneous systems, especially when responses are expected to vary smoothly among environmental gradients (19). However, the interpretation of GAM outputs requires careful consideration of spatial scale, data resolution, and potential sampling biases inherent to historical occurrence records (20; 21).

The Canary Islands archipelago represents an ideal natural laboratory to investigate long-term macroalgal diversity patterns under environmental heterogeneity. Located within the Canary Current upwelling system, the archipelago is characterized by a pronounced east-west environmental gradient driven by differential exposure to coastal upwelling from northwest Africa (22; 23; 24). Eastern islands are influenced by cooler, nutrient-rich waters, whereas western islands experience warmer, more oligotrophic conditions and greater water column stability. Previous studies have shown that this gradient shapes marine community composition across multiple taxa, including macroalgae, fishes, and invertebrates (1; 22; 25).

In addition to its strong environmental gradients, the Canary Islands host high levels of marine biodiversity and endemism and benefit from extensive historical biodiversity records spanning several decades (26). This combination of long-term biological data and spatially explicit environmental variability provides a robust framework for assessing macro-ecological patterns in macroalgal diversity at both island and regional scales. From a metacommunity perspective, macroalgal assemblages in the archipelago function as interconnected yet environmentally structured systems, where regional gradients can modulate local diversity, turnover, and persistence of species (10).

In this study, we integrate nearly three decades of macroalgal occurrence records with satellite-derived oceanographic data to examine spatiotemporal patterns in macroalgal diversity across the Canary Islands. Adopting a macro-ecological perspective, we characterize spatial patterns of diversity and species turnover along the east-west gradient, relate temporal and spatial variability in alpha, beta and gamma diversity to satellite-derived physical and biogeochemical variables across gamma diversity to satellite-derived physical and biogeochemical variables across seasonal to decadal scales, and assess long-term diversity trends over the past three decades in relation to environmental variability and regional heterogeneity.

By adopting a macro-ecological perspective grounded in long-term observational data, this study provides a regional-scale baseline for understanding how environmentally heterogeneous coastal systems respond to sustained climate variability, offering insights relevant for biodiversity assessment and adaptive management in warming oceans.

Methods

Study area

This study was conducted in the Canary Islands archipelago, located in the northeast Atlantic Ocean off the northwest coast of Africa. The archipelago comprises seven main islands (El Hierro, La Palma, La Gomera, Tenerife, Gran Canaria, Fuerteventura, and Lanzarote) that exhibit pronounced environmental heterogeneity due to their volcanic origin, oceanic setting, and exposure to the Canary Current system (22). A key feature of the region is a strong east-west environmental gradient driven by coastal upwelling off northwest Africa, which differentially affects the islands.

Based on this oceanographic context, islands were grouped into two biogeographic subregions for analytical purposes. The western group (El Hierro, La Palma, La Gomera, and Tenerife) is characterized by warmer, more oligotrophic waters and relatively stable oceanographic conditions, whereas the eastern group (Gran Canaria, Fuerteventura, and Lanzarote) is influenced by cooler, nutrient-enriched waters associated with upwelling processes (23; 27). This regional classification provided a consistent framework for exploring spatial differences in macroalgal diversity and its relationship with environmental variability.

Macroalgae records

Macroalgal occurrence records were compiled from herbarium collections and institutional biodiversity databases, primarily the Canary Islands Biodiversity Data Bank (CIBDB) (26). The dataset spans approximately three decades (1993–2023) and includes taxonomic identification at the species level, collection date (year and month), and geographic information (island and coordinates). In total, 64391 raw occurrence records were initially assembled.

For analytical robustness, we focused on four macroalgal orders that are both ecologically relevant and well represented throughout the study period: Bryopsidales, Ceramiales, Dictyotales, and Fucales. These groups include habitat-forming and structurally important taxa and provide sufficient temporal and spatial coverage for statistical analyses. After filtering, the dataset comprised 573 unique macroalgal species distributed across 49 orders and 111 families, sampled at 5437 distinct sites throughout the archipelago.

To reduce bias associated with uneven sampling effort across years and locations, duplicate records were removed by retaining a single occurrence per species, site, and year. This procedure minimized artificial inflation of species counts in intensively sampled areas or periods. The resulting dataset was aggregated at the island-year level and used to compute diversity metrics.

Diversity metrics

Macroalgal diversity was characterized using three complementary components: alpha, beta, and gamma diversity:

- Alpha diversity was assessed at the island level for each using two indices: species richness, defined as the number of distinct species recorded per island and year, and the Shannon diversity index, which incorporates both richness and relative species occurrence.
- Gamma diversity was calculated annually as the total number of unique macroalgal species recorded across all islands in a given year, providing a measure of regional-scale diversity over time.
- Beta diversity was used to quantify species turnover among islands. Presence-absence matrices were constructed for different time periods, and pairwise Jaccard dissimilarity indices were calculated between islands. These dissimilarities were used to examine the spatial structuring of macroalgal assemblages and to assess whether compositional similarity aligned with geographic proximity or regional environmental gradients.

Environmental data

Environmental variables were obtained from the Copernicus Marine Environmental Monitoring Service (CMEMS), including physical and biogeochemical reanalysis products for the Iberia-Biscay-Ireland region (28; 29; 30). Variables were selected based on their documented relevance to macroalgal physiology, productivity, and distribution in coastal systems.

The physical variables included sea surface temperature (SST, °C), potential temperature (thetao, °C), bottom temperature (bottomT, °C), sea surface salinity (SSS, PSU), mixed layer depth (MLD, m), mean sea level anomaly (zos, m), and zonal and meridional current components (uo and vo, m.s⁻¹) (29). Biogeochemical variables included chlorophyll-a concentration (Chl, mg.m⁻³), net primary production of carbon (nppv, mg, C, m⁻², day⁻¹), phytoplankton biomass expressed as carbon (phyc, mmol, C, m⁻³), dissolved inorganic carbon (dissic, mmol.m⁻³), dissolved oxygen (o2, μmol, kg⁻¹), seawater pH reported on the total scale (pH), surface partial pressure of CO₂ (spco2, atm), euphotic zone depth (zeu, m), and nutrient concentrations including ammonium (nh4, μM), nitrate (no3, μM), phosphate (po4, μM), silicate (si, μM), and dissolved iron (fe, mmol.m⁻³) (30).

All environmental datasets had a spatial resolution of 1/12° (approximately 8 km²) and monthly temporal resolution covering the period 1993-2023. For each macroalgal occurrence, environmental values were extracted from the nearest grid cell and matched to the corresponding month. Environmental variables were subsequently aggregated to island-level and regional-level time series using monthly, annual, seasonal, and decadal averages to align with biological data.

Data processing and statistical analysis

Biological and environmental datasets were integrated by matching macroalgal occurrences to concurrent environmental conditions in space and time. From the combined dataset, summary statistics were computed at multiple temporal scales (monthly, annual, and decadal) and spatial scales (island and regional).

Spatial patterns in alpha diversity were explored by comparing average diversity values across islands and by examining latitudinal and longitudinal gradients using grouped spatial bands. Temporal trends in diversity were assessed using annual and decadal time series for both island-level alpha diversity and regional gamma diversity. Seasonal variability in environmental drivers was examined by comparing monthly climatologies across decades.

To quantify relationships between macroalgal diversity and environmental variables, generalized additive models (GAMs) were fitted using the `mgcv` package in R (17). Species richness and Shannon diversity were used as response variables, with environmental predictors included as smooth terms to capture non-linear relationships. Models were fitted at three spatial resolutions:

- a single model for the archipelago,
- separate models for eastern and western regions, and
- island-specific models.

Model selection was guided by Akaike's Information Criterion (AIC) and statistical significance of smooth terms, with care taken to limit multicollinearity among predictors. Model performance was evaluated using explained deviance and adjusted coefficients of determination.

In parallel, a principal component analysis (PCA) was conducted on island-averaged environmental variables to synthesize dominant environmental gradients across the archipelago. PCA results were used to contextualize spatial patterns observed in diversity metrics and GAM outputs, particularly with respect to the east-west environmental gradient.

All analyses and visualization were performed in R (version 4.2) using packages *vegan*, *mgcv*, *ggplot2*, and *patchwork*. Statistical significance was evaluated at $\alpha = 0.05$.

Results

Spatial Distribution of macroalgal diversity.

Macroalgal diversity exhibited pronounced spatial heterogeneity across the Canary Islands. Total species richness differed markedly among islands, with Tenerife showing the highest number of recorded species, followed by La Palma, while El Hierro and La Gomera consistently displayed the lowest richness values (Fig.2). Gran Canaria, Fuerteventura, and Lanzarote occupied intermediate positions. These differences reflect a combination of environmental heterogeneity and uneven sampling intensity across islands (Tables:1, 4).

Beta diversity analyses revealed clear spatial structuring of macroalgal assemblages. Non-metric multidimensional scaling (NMDS) based on Jaccard dissimilarity showed distinct groupings among islands (Fig.5). La Gomera and El Hierro were positioned apart from the remaining islands, indicating more differentiated species composition, whereas Fuerteventura, Lanzarote, and Gran Canaria clustered closely together, reflecting higher floristic similarity. Tenerife and La Palma occupied intermediate positions in ordination space, sharing affinities with both eastern and western assemblages.

These spatial patterns indicate that macroalgal communities across the archipelago are not compositionally homogenous but instead structured along island-specific and regional gradients.

Temporal variability and long-term trends.

Macroalgal occurrence records spanned nearly three decades, although sampling effort varied strongly among years (Table:2). A pronounced peak in records occurred in 2005, which temporarily increased observed gamma diversity. After accounting for sampling normalization, decadal trends revealed a general increase in species richness during the 2000s, followed by a decline in the 2010s and 2020s at both island and regional scales.

Decadal trajectories of island-level species richness showed consistent patterns across the archipelago (Fig.3). Tenerife and La Palma maintained the highest absolute richness values throughout the study period, while El Hierro and La Gomera remained comparatively species-poor. Most islands exhibited a rise in richness during the 2000s, followed by a marked decrease in subsequent decades, particularly evident in the western islands.

At the regional scale, annual gamma diversity displayed interannual variability superimposed on these longer-term patterns, with richness peaks coinciding with periods of increased sampling but an overall reduction in recent decades.

Seasonal and inter-decadal variability of environmental drivers.

Environmental variables showed clear seasonal cycles and inter-decadal changes across the study period (Fig. 4). Sea surface temperature and surface partial pressure of CO_2 exhibited consistent increases across decades, while several biogeochemical variables displayed changes in the amplitude of their seasonal variability.

Nutrient-related variables, including nitrate, phosphate, and silicate concentration, showed pronounced winter maxima, although the magnitude of these peaks tended to diminish in more recent decades. Mixed layer depth also exhibited reduced seasonal amplitude, suggesting changes in water column stratification over time. In contrast, the bottom temperature increased slightly, particularly during the most recent decades.

These temporal patterns indicate coherent long-term changes in both physical and biogeochemical conditions across the archipelago.

Relationship between macroalgal diversity and environmental variables.

Statistical models identified strong associations between macroalgal diversity metrics and multiple environmental variables. Multiple linear regression analyses revealed significant relationships between species richness and variables such as sea surface temperature, current velocity, sea level anomaly, dissolved oxygen, and nutrient proxies (Table:6). Shannon diversity exhibited similar associations, with additional sensitivity to phosphate and silicate concentrations.

Generalized additive models (GAMs) provided a more detailed characterization of these relationships by capturing non-linear responses (Tables: 7, 8, 9, 12, 13). For species richness, the best-performing Poisson GAM explained approximately 89% of the deviance, with significant smooth terms for temperature-related variables, hydrodynamic indicators, and biogeochemical

parameters. Richness tended to increase across intermediate ranges of temperature and oxygen concentration, while extreme values of stratification-related variables were associated with lower richness.

For Shannon diversity, GAMs explained approximately 65% of the deviance, with significant effects of sea surface temperature, chlorophyll-a concentration, dissolved oxygen, phosphate, and silicate. Non-linear responses indicated that evenness varied across environmental gradients rather than responding monotonically to single drivers.

Model performance improved when the analyses were stratified by region, with separate GAMs for eastern and western islands yielding higher explanatory power compared to a single archipelago-wide model.

Environmental gradients and multivariate structure.

Principal component analysis (PCA) of environmental variables summarized the dominant gradients across the archipelago (Tables:11, 14). The first principal component (PC1) accounted for approximately 31.5% of the total variance and clearly separated eastern and western islands. PC1 contrasted warmer, more oligotrophic, higher- CO_2 conditions with cooler, nutrient-enriched environments.

This multivariate environmental gradient aligned with observed spatial patterns in macroalgal diversity and composition. Eastern islands were associated with higher nutrient availability and greater short-term variability, whereas western islands occupied regions of the environmental space characterized by warmer and more stable conditions.

Together, these results indicate that macroalgal diversity patterns across the Canary Islands are structured by persistent spatial gradients and long-term environmental variability, as captured by both univariate and multivariate analyses.

Discussion

This study provides a long-term, macro-ecological assessment of macroalgal diversity patterns in the Canary Islands by integrating historical occurrence records with satellite-derived environmental data spanning nearly three decades. The results reveal consistent spatial and temporal structuring of macroalgal diversity across the archipelago, closely aligned with persistent environmental gradients and long-term variability in the oceanographic conditions.

Spatial structuring and the east-west environmental gradient.

One of the most robust outcomes of this study is the clear spatial differentiation of macroalgal communities among islands, structured along the documented east-west environmental gradient of the Canary archipelago. Differences in alpha diversity among islands (Fig.2) and the compositional clustering revealed by beta diversity analysis (Fig.5) indicate that macroalgal assemblages are not uniformly distributed across the archipelago. Instead, they reflect regional environmental heterogeneity associated with differential exposure to coastal upwelling and oceanographic conditions (22; 23).

The multivariate environmental space summarized by the principal component analysis reinforces this interpretation. The PC1 captured a dominant gradient separating cooler, nutrient-enriched eastern islands from warmer, more oligotrophic western islands (Tables:11, 14). The alignments between this environmental gradient and spatial patterns in macroalgal diversity support previous observations that regional oceanography plays a central role in shaping marine communities in the Canary Islands (1; 25; 27). From a macro-ecological perspective, these results highlight how persistent environmental gradients can structure metacommunity patterns at relatively small spatial scales within an archipelago.

Temporal variability and long-term diversity trends.

Temporal analyses revealed coherent decadal patterns in macroalgal diversity across islands, characterized by a generalized increase in species richness during the 2000s decade followed by a decline in more recent decades (Fig.3). Although interannual variability and uneven sampling efforts complicate direct interpretation of temporal trends (Table:2), the consistency of these patterns across multiple islands suggests that they are not only driven by sampling tools.

At the regional scale, gamma diversity exhibited marked interannual fluctuations imposed on longer-term changes. These temporal patterns occurred in parallel with inter-decadal changes in environmental drivers, including increases in sea surface temperature and surface CO_2 levels and reductions in the seasonal amplitude of nutrient-related variables (Fig.4). While the observational nature of the data precludes causal attribution, the temporal coincidence between diversity changes and environmental variability is consistent with broader evidence of climate-related reorganization in marine communities (3; 15).

Associations between diversity metrics and environmental drivers.

Statistical modeling indicated that macroalgal diversity metrics were associated with multiple physical and biogeochemical variables rather than responding to a single dominant driver. Both multiple linear regressions and generalized additive models identified temperature-related variables, hydrodynamic indicators, and nutrient proxies as significant correlates of species richness and Shannon diversity (Tables:6, 7, 8, 9, 12, 13). The non-linear relationships captured by GAMs underscore the

importance of considering complex and potentially interacting environmental gradients when examining long-term ecological patterns.

Notably, the model performance improved when analyses were stratified by region, with separate models for eastern and western islands explaining a greater proportion of variance than archipelago-wide models. These results suggest that macroalgal-environmental relationships are context-dependent and that aggregating environmentally distinct regions may obscure meaningful patterns. Similar scale-dependent responses have been documented in other heterogeneous marine systems, where regional stratification improves the detection of ecological signals (18; 31).

Signals compatible with community reorganization.

The observed spatial and temporal patterns are compatible with a gradual reorganization of macroalgal communities across the archipelago. In particular, the decline in richness observed in western islands during recent decades (Fig.3), combined with their position in warmer and more oligotrophic regions of the environmental space 11, 14, is consistent with reported reductions in structurally complex temperate macroalgae in these areas (25; 26). Conversely, eastern islands, influenced by nutrient enrichment associated with upwelling, maintained comparatively higher richness and compositional similarity (Figs.2, 3).

These patterns align with the concept of incipient tropicalization described in other marine systems, where warming conditions favor warm-affinity or opportunistic taxa while structurally complex temperate species decline (13; 32). However, it is important to emphasize that the present study does not directly quantify changes in species' thermal affinities or functional traits. Therefore, tropicalization is interpreted here as a descriptive framework consistent with observed patterns rather than as a demonstrated mechanistic process.

Methodological considerations and limitations

Several limitations should be considered when interpreting these results. First, macroalgal occurrence records are derived from opportunistic and non-standardized sampling, resulting in uneven sampling efforts across years and islands (Table:2). Although filtering and aggregation procedures were applied to reduce bias, absolute changes in abundance or biomass cannot be inferred. Second, environmental variables were extracted at a spatial resolution of approximately $8km^2$, which may obscure fine-scale habitat heterogeneity within islands, such as depth gradients or localized wave exposure.

Additionally, the models included only abiotic environmental variables. Biotic interactions, such as herbivory, competition, and the influence of invasive species, are known to affect macroalgal dynamics, but were not explicitly accounted for (10; 12). As a result, the associations identified here should be interpreted as macro-ecological correlations rather than comprehensive explanations of community dynamics.

Implications for regional-scale assessments.

Despite these limitations, the integration of long-term biological records with satellite-derived environmental data provides valuable insights into how macroalgal diversity is structured across environmentally heterogeneous coastal systems. This consistency between spatial gradients, temporal trends, and multivariate environmental structure highlights the utility of macro-ecological approaches for detecting broad-scale ecological patterns under climate variability.

By identifying regions that exhibit consistent diversity trajectories and environmental context, this study offers a foundation for future research aimed at disentangling mechanisms, assessing functional consequences, and informing regionally differentiated conservation strategies in the Canary Islands and similar archipelagos.

Conclusions

This study integrates nearly three decades of macroalgal occurrence records with satellite-derived environmental data to provide a regional-scale assessment of macroalgal diversity patterns in the Canary Islands. By adopting a macro-ecological perspective, our analyses reveal that macroalgal diversity and community composition across the archipelago are strongly structured by persistent spatial gradients and long-term environmental variability.

Across islands, marked differences in alpha and beta diversity demonstrate that macroalgal communities are not uniformly distributed but instead reflect island-specific and regional environmental context (Figs.2, 5). These partial patterns align with the pronounced east-west environmental gradient characteristic of the Canary Islands, as summarized by multivariate analyses of physical biogeochemical variables (Tables:11, 14). Eastern islands, influenced by nutrient-enriched waters associated with upwelling, and western islands characterized by warmer and more oligotrophic conditions, occupy distinct regions of the environmental space and exhibit contrasting diversity patterns.

Temporal analyses further indicate that macroalgal diversity has not remained static over the past three decades. Decadal trajectories show a generalized increase in species richness during the 2000s followed by a decline in more recent decades, particularly in western islands (Fig.3). These changes occurred alongside coherent inter-decadal trends in environmental drivers,

including increases in sea surface temperature and modifications in nutrient seasonality (Fig4). While the observational nature of the data precludes causal attribution, the temporal correspondence between diversity patterns and environmental variability is consistent with broader evidence of climate-related reorganization in marine ecosystems.

Statistical models demonstrated that macroalgal diversity metrics are associated with multiple environmental variables, with non-linear relationships captured by generalized additive models (Tables:6, 7, 8, 9, 12, 13). Importantly, model performance improved when analyses were stratified by region, highlighting the value of accounting for environmental heterogeneity when examining diversity–environment relationships in complex coastal systems.

Taking all into account, these results are compatible with a gradual reorganization of macroalgal communities under contemporary environmental variability, including signals consistent with incipient changes in community composition across the archipelago. However, these patterns should be interpreted as macro-ecological associations rather than direct evidence of underlying mechanisms, given the limitations inherent to historical occurrence data and coarse-resolution environmental layers.

By demonstrating the utility of integrating long-term biodiversity records with satellite-derived environmental information, this study provides a robust baseline for tracking macroalgal diversity dynamics in environmentally heterogeneous regions. The approach and findings presented here contribute to regional-scale biodiversity assessments and offer a framework for future studies aimed at refining mechanistic understanding, improving predictive capacity, and supporting adaptive management strategies in coastal ecosystems facing sustained environmental change.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the author(s) used ChatGPT for language editing and text refinement. After using this tool, the author(s) carefully reviewed and edited all generated material and take full responsibility for the content, accuracy, and integrity of the published article. The use of this tool did not replace human judgment, critical analysis, or scientific interpretation.

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Author contributions statement

E.G. conceived the idea, and analyze the results, figures and first draft, D.G-A. Revise codes and artwork, and provide feedback on the first draft. R.R: Revise the whole manuscript, and polish the final version of this piece. All authors reviewed the manuscript.

Figures and Tables

To include, in this order: **Accession codes** (where applicable); **Competing interests** (mandatory statement).

The corresponding author is responsible for submitting a [competing interests statement](#) on behalf of all authors of the paper. This statement must be included in the submitted article file.

Figures

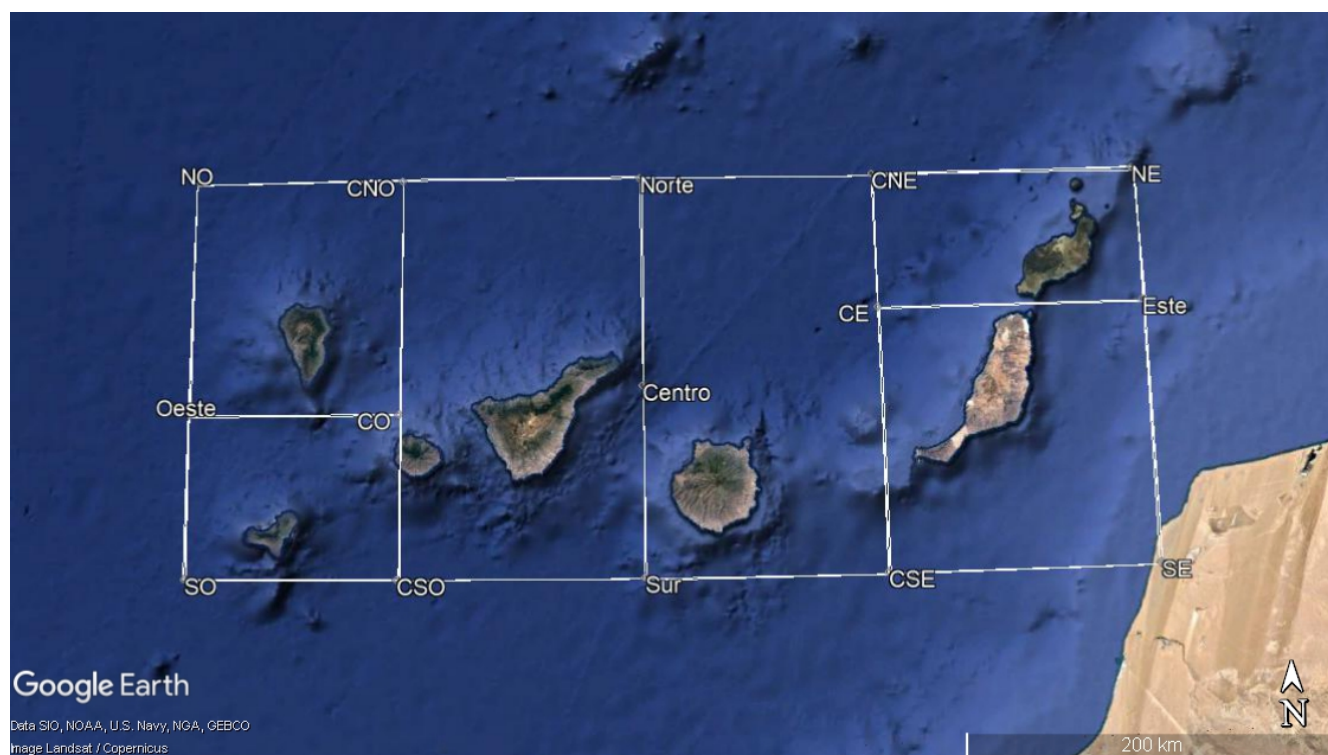


Figure 1. Study area and regional classification: Map of the Canary Islands archipelago showing the location of the seven study islands and their classification into western and eastern regions. The schematic representation highlights the main oceanographic features influencing the east–west environmental gradient.

Tables

1 Annex I: Supplementary Resources

Table 1. Taxonomic breadth and temporal extent of the macroalgal occurrence dataset used in this study: This table summarizes the number of taxa recorded at each hierarchical level, the number of islands included, and the total temporal coverage of the records.

Level	Count
Reinos	2
Filos	4
Clases	14
Órdenes	49
Familias	111
Géneros	289
Especies	573
Islas	7
Años	30

Table 2. Number of macroalgal occurrence records compiled per year between 1992 and 2021: This table illustrates the uneven sampling effort across years, which was accounted for during data processing and interpretation.

Year	Records
1992	714
1993	1615
1994	463
1995	248
1996	2602
1997	317
1998	1396
1999	749
2000	846
2001	2221
2002	1504
2003	1589
2004	707
2005	29986
2006	290
2007	426
2008	7319
2009	306
2010	59
2011	2797
2012	343
2013	163
2014	6021
2015	581
2016	62
2017	108
2018	418
2019	55
2020	43
2021	442

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Table 3. Most frequently recorded macroalgal species: List of the fifteen macroalgal species with the highest number of occurrence records in the dataset, indicating their relative representation across the study period and islands.

Species	Records
<i>Gongolaria abiesmarina</i>	3874
<i>Dunaliella salina</i>	1704
<i>Gelidium canariense</i>	1515
<i>Lobophora variegata</i>	1481
<i>Caulerpa prolifera</i>	1475
<i>Alsidium corallinum</i>	1341
<i>Padina pavonica</i>	1334
<i>Fucus spiralis</i>	1332
<i>Caulerpa racemosa</i>	1258
<i>Lophocladia trichoclados</i>	1168
<i>Halopteris scoparia</i>	1112
<i>Caulerpa webbiana</i>	1088
<i>Jania pedunculata</i>	1061
<i>Corallina ferreyrae</i>	1056
<i>Cystoseira compressa</i>	1014

Table 4. Distribution of macroalgal records by island: Total number of macroalgal occurrence records compiled for each island in the Canary Islands archipelago, reflecting spatial differences in sampling intensity.

Islad	Records
Lanzarote	20756
Tenerife	8822
Gran Canaria	8794
La Palma	8566
El Hierro	8343
Fuerteventura	7515
La Gomera	1594

Table 5. Summary statistics of environmental variables: Descriptive statistics of the physical and biogeochemical environmental variables extracted from Copernicus Marine Environment Monitoring Service products and matched to macroalgal records. Values correspond to monthly data aggregated across the study period (29; 30).

Variable	Analysis name	Unit	Min	Q1	Median	Mean	Q3	Max	NA
Sea Surface Temperature	SST	°C	18.92	20.31	20.47	20.63	20.70	23.35	1846
Bottom temperature	bottomT	°C	12.04	13.67	13.92	13.82	14.01	15.36	9157
Mixed Layer Depth	MLD	m	10.80	16.40	17.80	19.08	21.50	37.20	1846
Sea Surface Salinity	SSS	PSU	36.25	36.55	36.59	36.69	36.85	37.22	1846
Potential temperature	thetao	°C	18.92	20.31	20.47	20.63	20.70	23.35	1846
Zonal current component	uo	m s ⁻¹	-0.2820	-0.1230	-0.1000	-0.0983	-0.0770	0.1600	1846
Meridional current component	vo	m s ⁻¹	-0.2680	-0.1410	-0.1100	-0.1002	-0.0630	0.2560	1846
Mean sea level anomaly	zos	m	-0.2940	-0.2170	-0.2100	-0.2052	-0.1930	-0.0800	1846
Chlorophyll-a concentration	Chl	mg m ⁻³	0.050	0.070	0.070	0.087	0.080	0.280	1846
Dissolved inorganic carbon	dissic	mmol m ⁻³	2.140	2.170	2.170	2.172	2.180	2.190	1846
Mole concentration of iron	fe	mmol m ⁻³	0.0003	0.0007	0.0007	0.0008	0.0009	0.0016	1846
Mole concentration of ammonium	nh4	μM	0.0	0.0	0.0	0.0	0.0	0.1	1846
Mole concentration of nitrate	no3	μM	0.0000	0.0000	0.0000	0.0006	0.0000	0.1000	1846
Net primary production of carbon	nppv	mg C m ⁻² day ⁻¹	0.000	1.000	1.000	1.779	1.000	20.000	1846
Mole concentration of dissolved oxygen	o2	μmol kg ⁻¹	224.0	233.0	234.0	233.6	235.0	242.0	1846
Sea water pH (total scale)	ph	—	8.011	8.048	8.050	8.050	8.052	8.074	1846
Phytoplankton (as carbon)	phyc	mmol C m ⁻³	0.8000	0.9000	1.0000	0.9952	1.0000	1.7000	1846
Mole concentration of phosphate	po4	μM	0.0300	0.0400	0.0400	0.0471	0.0500	0.1100	1846
Mole concentration of silicate	si	μM	0.700	1.200	1.300	1.284	1.300	1.900	1846
Surface partial pressure of CO ₂	spco2	atm	37.30	39.60	40.00	39.96	40.22	44.38	1846
Euphotic zone depth	zeu	m	61.60	89.70	92.40	93.08	96.50	135.60	1846

Table 6. Multiple linear regression models for diversity metrics. Results of multiple linear regression models relating macroalgal species richness and Shannon diversity to selected environmental variables. Estimates, standard errors, test statistics, and p-values are reported. Significant predictors are highlighted.

Term	Estimate	Std. Error	t value	p value
Richness				
(Intercept)	-2841.00	849.00	-3.35	0.00088
SST_mean	35.80	8.67	4.13	0.00004
bottomT_mean	4.28	2.74	1.56	0.119
uo_mean	169.00	25.80	6.56	1.45e-10
vo_mean	-118.00	29.90	-3.95	0.00009
zos_mean	243.00	67.80	3.58	0.00039
dissic_mean	857.00	257.00	3.33	0.00092
fe_mean	-22648.00	8278.00	-2.74	0.00646
nppv_mean	1.64	0.94	1.74	0.0828
o2_mean	3.60	1.71	2.11	0.0354
phyc_mean	47.80	13.90	3.44	0.00063
po4_mean	-259.00	153.00	-1.69	0.0914
spco2_mean	-14.50	2.87	-5.04	6.68e-07
zeu_mean	-0.23	0.16	-1.42	0.155
Shannon				
(Intercept)	-91.30	25.80	-3.53	0.00045
SST_mean	1.01	0.15	6.73	5.12e-11
bottomT_mean	0.26	0.14	1.89	0.0590
uo_mean	7.50	1.34	5.59	3.90e-08
vo_mean	-3.41	1.51	-2.25	0.0248
zos_mean	8.38	3.43	2.44	0.0150
chl_mean	-8.31	5.66	-1.47	0.143
dissic_mean	47.80	12.20	3.92	0.00010
phyc_mean	2.97	1.20	2.48	0.0135
po4_mean	-20.80	7.51	-2.76	0.00595
si_mean	0.95	0.47	2.02	0.0436
spco2_mean	-0.84	0.14	-6.22	1.16e-09
zeu_mean	-0.02	0.01	-2.70	0.00722

Table 7. Generalized additive models for macroalgal diversity. Summary of generalized additive models fitted to species richness and Shannon diversity, including estimated degrees of freedom, test statistics, and p-values for each smooth term.

Smooth term	edf	Ref.df	F	p value
Richness				
s(SST_mean)	2.82	9	0.32	0.00062
s(bottomT_mean)	5.41	9	1.76	0.00333
s(mlotst_mean)	6.14	9	2.90	4.69e-05
s(so_mean)	4.71	9	0.83	0.0866
s(thetao_mean)	2.82	9	0.32	0.00062
s(uo_mean)	8.78	9	2.81	0.00141
s(vo_mean)	0.10	9	0.01	0.234
s(zos_mean)	8.26	9	3.14	0.00016
s(chl_mean)	5.83	9	2.84	1.06e-05
s(dissic_mean)	3.26	9	0.57	0.124
s(fe_mean)	3.46	9	0.96	0.0191
s(nppv_mean)	5.26	9	3.07	3.42e-06
s(o2_mean)	2.81	9	1.17	0.00245
s(ph_mean)	7.43	9	3.77	1.30e-06
s(phyc_mean)	4.62	9	1.31	0.0101
s(po4_mean)	3.99	9	1.24	0.00840
s(si_mean)	4.85	9	3.14	1.99e-06
s(spcO2_mean)	7.95	9	3.86	1.10e-06
s(zeu_mean)	0.00	9	0.00	0.882
Shannon				
s(SST_mean)	4.08	9	0.21	0.00239
s(bottomT_mean)	3.96	9	1.99	0.00029
s(mlotst_mean)	5.18	9	1.66	0.00487
s(so_mean)	0.00	9	0.00	0.811
s(thetao_mean)	4.08	9	0.21	0.00239
s(uo_mean)	0.18	9	0.04	0.168
s(vo_mean)	0.00	9	0.00	0.373
s(zos_mean)	7.81	9	2.13	0.00555
s(chl_mean)	6.31	9	3.30	1.84e-06
s(dissic_mean)	2.52	9	0.42	0.158
s(fe_mean)	3.42	9	1.18	0.00699
s(nppv_mean)	6.30	9	2.89	3.88e-05
s(o2_mean)	7.69	9	5.09	<2e-16
s(ph_mean)	5.25	9	1.89	0.00050
s(phyc_mean)	2.79	9	0.90	0.0121
s(po4_mean)	5.99	9	2.55	0.00022
s(si_mean)	4.58	9	3.12	1.74e-06
s(spcO2_mean)	7.71	9	2.69	2.73e-05
s(zeu_mean)	5.51	9	1.72	0.00375

Table 8. Poisson GAM for species richness. Results of the generalized additive model fitted to species richness using a Poisson distribution. Reported values include estimated degrees of freedom, test statistics, and p-values for each environmental predictor.

Smooth term	edf	Ref.df	Chi.sq	p value
s(SST_mean)	4.50	9	0.004	<2e-16
s(bottomT_mean)	7.76	9	245.93	<2e-16
s(mlotst_mean)	8.22	9	203.46	<2e-16
s(so_mean)	8.41	9	76.22	<2e-16
s(thetao_mean)	4.50	9	0.004	<2e-16
s(uo_mean)	9.00	9	168.76	<2e-16
s(vo_mean)	7.02	9	79.56	<2e-16
s(zos_mean)	8.64	9	136.16	<2e-16
s(chl_mean)	8.93	9	114.34	<2e-16
s(dissic_mean)	8.84	9	120.80	<2e-16
s(fe_mean)	8.87	9	103.57	<2e-16
s(nppv_mean)	7.78	9	87.99	<2e-16
s(o2_mean)	7.16	9	170.49	<2e-16
s(ph_mean)	8.52	9	250.35	<2e-16
s(phyc_mean)	9.00	9	196.73	<2e-16
s(po4_mean)	8.83	9	110.62	<2e-16
s(si_mean)	7.93	9	109.83	<2e-16
s(spcO2_mean)	9.00	9	294.07	<2e-16
s(zeu_mean)	8.89	9	86.45	<2e-16

Table 9. Gaussian GAM for Shannon diversity. Results of the generalized additive model fitted to the Shannon diversity index using a Gaussian distribution. Reported values include estimated degrees of freedom, F-statistics, and p-values.

Smooth term	edf	Ref.df	F	p value
s(SST_mean)	4.08	9	0.21	0.0024
s(bottomT_mean)	3.96	9	1.99	0.00029
s(mlotst_mean)	5.18	9	1.66	0.0049
s(so_mean)	0.00	9	0.00	0.811
s(thetao_mean)	4.08	9	0.21	0.0024
s(uo_mean)	0.18	9	0.04	0.168
s(vo_mean)	0.00	9	0.00	0.373
s(zos_mean)	7.81	9	2.13	0.0055
s(chl_mean)	6.31	9	3.30	1.8e-06
s(dissic_mean)	2.52	9	0.42	0.158
s(fe_mean)	3.42	9	1.18	0.0070
s(nppv_mean)	6.30	9	2.89	3.9e-05
s(o2_mean)	7.69	9	5.09	<2e-16
s(ph_mean)	5.25	9	1.89	0.00050
s(phyc_mean)	2.79	9	0.90	0.012
s(po4_mean)	5.99	9	2.55	0.00022
s(si_mean)	4.58	9	3.12	1.7e-06
s(spcO2_mean)	7.71	9	2.69	2.7e-05
s(zeu_mean)	5.51	9	1.72	0.0038

Table 10. GAM fitted to visualization subset. Generalized additive model for species richness fitted to a reduced visualization subset of the data. Model performance metrics and smooth term statistics are reported.

Smooth term	edf	Ref.df	Chi.sq	p value
s(SST_mean)	2.8000	9	3.302	< 2e-16
s(bottomT_mean)	9.0000	9	54.722	< 2e-16
s(mlotst_mean)	9.0000	9	41.813	< 2e-16
s(so_mean)	6.0019	9	17.781	8.73e-05
s(thetao_mean)	2.8000	9	3.302	< 2e-16
s(uo_mean)	8.7343	9	32.021	< 2e-16
s(vo_mean)	8.8313	9	39.450	< 2e-16
s(zos_mean)	8.6311	9	50.598	< 2e-16
s(chl_mean)	1.2697	9	4.539	0.00606
s(dissic_mean)	0.9321	8	10.449	1.86e-05
s(fe_mean)	8.9929	9	55.055	< 2e-16
s(no3_mean)	2.9999	3	4.619	0.06180
s(nppv_mean)	6.2908	9	29.497	< 2e-16
s(o2_mean)	5.0230	9	20.658	< 2e-16
s(ph_mean)	5.8639	9	27.064	< 2e-16
s(phyc_mean)	7.1125	9	19.656	8.64e-05
s(po4_mean)	5.5078	9	22.559	4.30e-07
s(si_mean)	6.6748	9	45.417	< 2e-16
s(spc02_mean)	7.5839	9	22.537	< 2e-16
s(zeu_mean)	6.6533	9	39.218	< 2e-16

Table 11. Principal component analysis summary. Eigenvalues, proportion of variance explained, and cumulative variance for each principal component derived from the environmental variables used in the analysis.

Component	Standard deviation	Proportion of Variance	Cumulative Proportion
PC1	2.5116	0.31540	0.31540
PC2	1.7433	0.15200	0.46740
PC3	1.6886	0.14260	0.60990
PC4	1.4376	0.10330	0.71330
PC5	1.08115	0.05844	0.77171
PC6	1.03818	0.05389	0.82560
PC7	0.94352	0.04451	0.87012
PC8	0.88598	0.03925	0.90936
PC9	0.68537	0.02349	0.93285
PC10	0.61112	0.01867	0.95152
PC11	0.51618	0.01332	0.96485
PC12	0.47498	0.01128	0.97613
PC13	0.40112	0.00804	0.98417
PC14	0.35934	0.00646	0.99063
PC15	0.30186	0.00456	0.99518
PC16	0.26113	0.00341	0.99859
PC17	0.14010	0.00098	0.99957
PC18	0.08751	0.00038	0.99996
PC19	0.02950	0.00004	1.00000
PC20	4.285e-16	0.00000	1.00000

Table 12. Regional GAM for species richness. Generalized additive model results for species richness fitted at the regional scale. Estimated degrees of freedom, test statistics, and p-values are reported for each smooth term.

Smooth term	edf	Ref.df	Chi.sq	p-value
s(SST_mean)	4.500	9	0.004	< 2e-16
s(bottomT_mean)	7.760	9	245.930	< 2e-16
s(mlotst_mean)	8.224	9	203.457	< 2e-16
s(so_mean)	8.407	9	76.222	< 2e-16
s(thetao_mean)	4.500	9	0.004	< 2e-16
s(uo_mean)	8.998	9	168.761	< 2e-16
s(vo_mean)	7.020	9	79.559	< 2e-16
s(zos_mean)	8.636	9	136.156	< 2e-16
s(chl_mean)	8.934	9	114.340	< 2e-16
s(dissic_mean)	8.844	9	120.795	< 2e-16
s(fe_mean)	8.869	9	103.565	< 2e-16
s(nppv_mean)	7.777	9	87.989	< 2e-16
s(o2_mean)	7.157	9	170.490	< 2e-16
s(ph_mean)	8.515	9	250.347	< 2e-16
s(phyc_mean)	9.000	9	196.731	< 2e-16
s(po4_mean)	8.825	9	110.617	< 2e-16
s(si_mean)	7.927	9	109.826	< 2e-16
s(spcO2_mean)	9.000	9	294.070	< 2e-16
s(zeu_mean)	8.893	9	86.448	< 2e-16

Table 13. Regional GAM for Shannon diversity:Generalized additive model results for Shannon diversity fitted at the regional scale, reporting smooth term statistics and overall model performance.

Smooth term	edf	Ref.df	F	p-value
s(SST_mean)	4.081	9	0.210	0.00239 **
s(bottomT_mean)	3.963	9	1.994	0.00029 ***
s(mlotst_mean)	5.177	9	1.661	0.00487 **
s(so_mean)	0.000	9	0.000	0.81118
s(thetao_mean)	4.081	9	0.210	0.00239 **
s(uo_mean)	0.184	9	0.035	0.16844
s(vo_mean)	0.000	9	0.000	0.37314
s(zos_mean)	7.808	9	2.130	0.00555 **
s(chl_mean)	6.310	9	3.302	1.8e-06 ***
s(dissic_mean)	2.516	9	0.416	0.15787
s(fe_mean)	3.421	9	1.176	0.00699 **
s(nppv_mean)	6.295	9	2.885	3.9e-05 ***
s(o2_mean)	7.692	9	5.087	< 2e-16 ***
s(ph_mean)	5.254	9	1.892	0.00050 ***
s(phyc_mean)	2.792	9	0.895	0.01212 *
s(po4_mean)	5.991	9	2.545	0.00022 ***
s(si_mean)	4.576	9	3.120	1.7e-06 ***
s(spcO2_mean)	7.709	9	2.687	2.7e-05 ***
s(zeu_mean)	5.507	9	1.716	0.00375 **

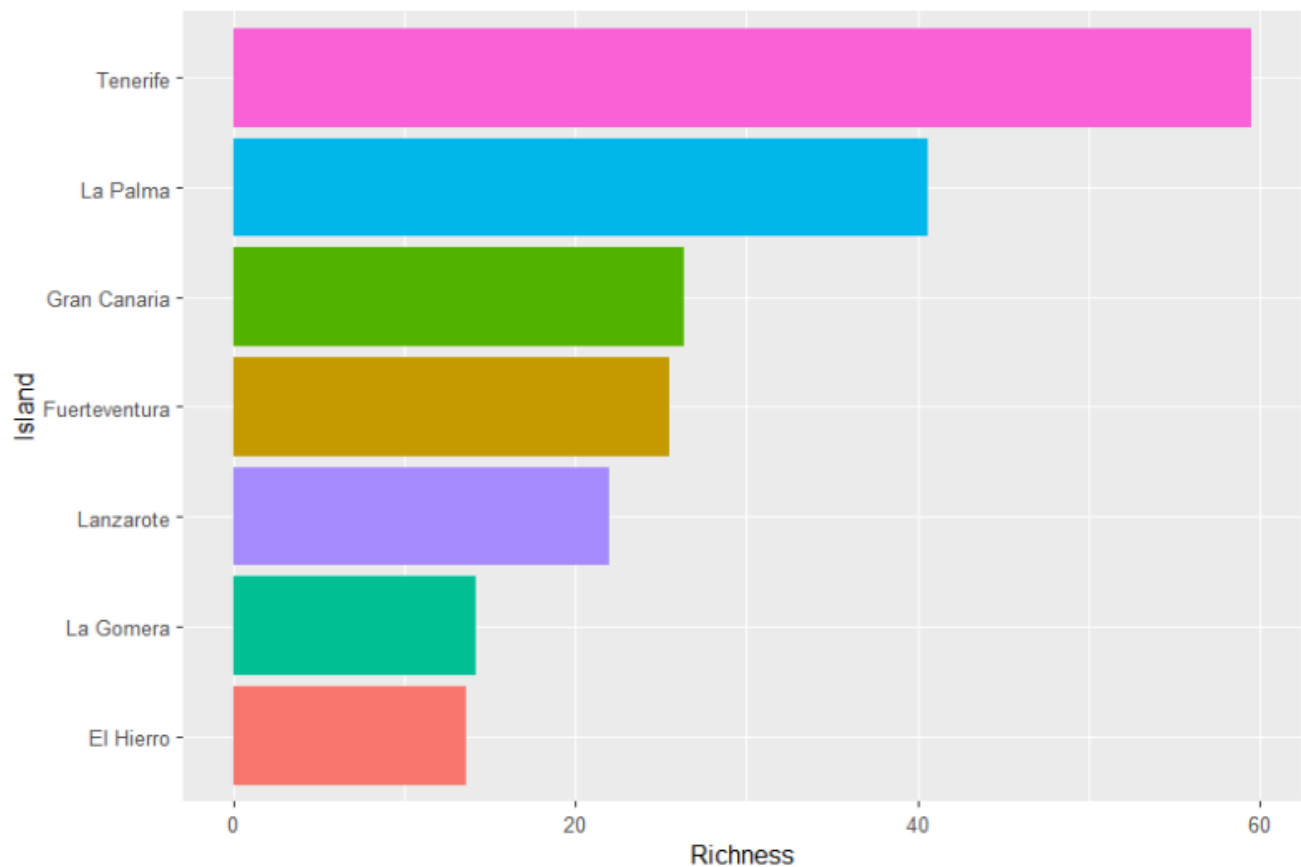


Figure 2. Spatial distribution of macroalgal species richness: Total macroalgal species richness recorded for each island across the study period. Differences among islands reflect spatial heterogeneity and variation in sampling intensity

Table 14. PCA loadings of environmental variables: Loadings of environmental variables on principal components derived from the principal component analysis, indicating the contribution of each variable to the main environmental gradients.

Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11	PC12	PC13	PC14	PC15	PC16	PC17	PC18	PC19
s(SST_mean)	0.351	-0.348	-0.096	0.117	0.112	0.210	0.078	0.085	-0.041	0.097	-0.081	-0.075	0.186	-0.051	-0.144	-0.283	-0.036	-0.011	-0.707
s(bottomT_mean)	-0.075	0.052	0.048	-0.090	0.483	-0.351	0.380	0.302	0.111	-0.040	-0.072	0.522	0.286	0.064	0.043	-0.083	0.042	-0.021	0.000
s(mlotst_mean)	0.125	-0.029	0.185	0.136	0.310	-0.146	0.352	-0.660	-0.139	-0.251	0.302	-0.194	0.016	0.171	-0.109	0.034	-0.025	-0.025	0.000
s(so_mean)	0.151	-0.020	0.134	-0.261	-0.488	0.283	0.153	-0.112	-0.121	-0.092	-0.028	0.320	0.428	0.433	-0.009	0.195	0.028	0.039	0.000
s(thetao_mean)	0.351	-0.348	-0.096	0.117	0.112	0.210	0.078	0.085	-0.041	0.097	-0.081	-0.075	0.186	-0.051	-0.144	-0.283	-0.036	-0.011	0.707
s(uo_mean)	0.132	0.030	0.338	0.261	-0.230	-0.174	0.059	0.470	-0.008	-0.192	0.512	-0.247	0.280	-0.182	0.076	0.098	-0.058	0.026	0.000
s(vo_mean)	0.162	0.135	0.288	0.298	-0.259	-0.144	0.328	0.143	-0.018	-0.157	-0.512	0.004	-0.367	0.036	-0.377	0.019	0.037	-0.028	0.000
s(zos_mean)	0.223	0.077	0.037	-0.501	0.093	-0.217	-0.164	0.050	0.020	0.220	0.136	-0.084	0.040	-0.109	-0.668	0.258	-0.064	-0.037	0.000
s(chl_mean)	-0.209	-0.448	0.311	-0.094	-0.063	-0.107	-0.015	-0.058	-0.080	0.130	-0.008	-0.016	-0.002	-0.141	-0.006	0.017	0.761	-0.092	0.000
s(dissic_mean)	0.174	0.103	-0.137	-0.416	-0.101	-0.328	0.288	0.054	-0.133	0.093	-0.276	-0.530	0.099	0.063	0.392	-0.103	0.038	-0.031	0.000
s(fe_mean)	0.180	-0.106	0.029	0.295	-0.074	-0.475	-0.416	-0.107	0.368	0.208	-0.075	-0.020	0.132	0.495	0.043	-0.044	0.015	-0.025	0.000
s(nppv_mean)	0.238	0.360	-0.337	-0.006	-0.043	0.080	0.060	0.162	-0.013	-0.051	0.383	0.058	-0.260	0.292	-0.095	-0.303	0.495	-0.104	0.000
s(o2_mean)	-0.273	0.320	0.242	-0.026	-0.241	-0.063	-0.004	-0.227	0.001	0.225	0.031	0.035	0.252	-0.180	-0.178	-0.676	-0.098	-0.055	0.000
s(ph_mean)	0.379	0.074	0.345	-0.116	0.075	0.060	-0.188	-0.051	-0.083	0.026	0.000	0.219	-0.178	-0.125	0.301	-0.025	-0.112	-0.682	0.000
s(phyc_mean)	0.193	0.413	-0.073	0.142	0.145	0.144	-0.184	-0.165	0.229	-0.256	-0.293	-0.100	0.437	-0.316	-0.007	0.172	0.360	-0.012	0.000
s(po4_mean)	-0.201	-0.044	0.266	-0.237	0.258	0.136	-0.344	0.224	-0.086	-0.531	-0.142	-0.244	0.021	0.354	-0.117	-0.260	-0.025	-0.020	0.000
s(si_mean)	0.064	-0.279	-0.143	-0.281	-0.264	-0.111	0.105	-0.126	0.619	-0.456	0.087	0.089	-0.131	-0.224	0.002	-0.172	-0.065	-0.049	0.000
s(spc2_mean)	-0.386	-0.075	-0.325	0.160	-0.078	0.025	0.166	0.073	0.047	-0.017	-0.014	-0.191	0.212	0.119	-0.216	0.152	-0.056	-0.709	0.000
s(zeu_mean)	-0.062	0.113	0.344	-0.074	0.176	0.421	0.268	0.078	0.582	0.345	0.052	-0.245	-0.084	0.189	0.060	0.080	0.017	0.013	0.000

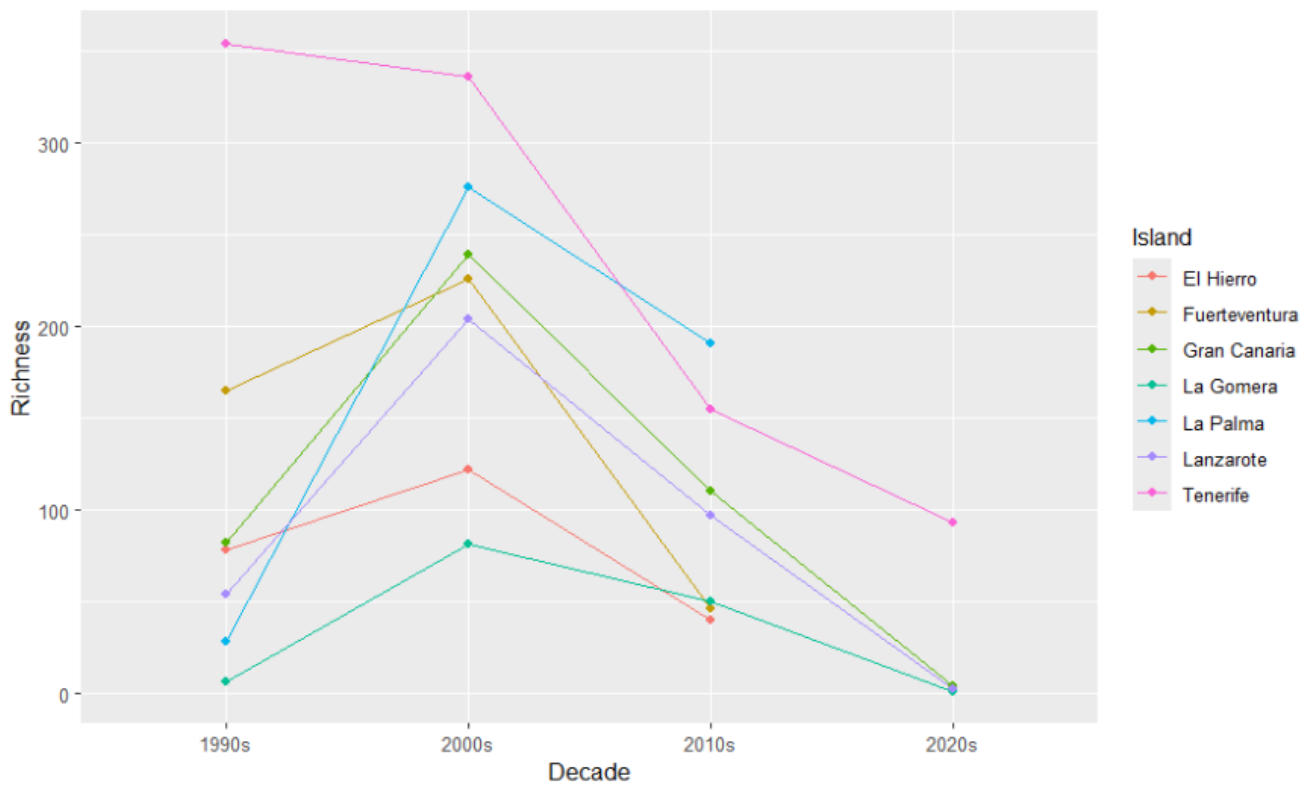


Figure 3. Decadal changes in macroalgal species richness: Decadal averages of macroalgal species richness by island from the 1990s to the 2020s, illustrating long-term temporal variability across the archipelago.

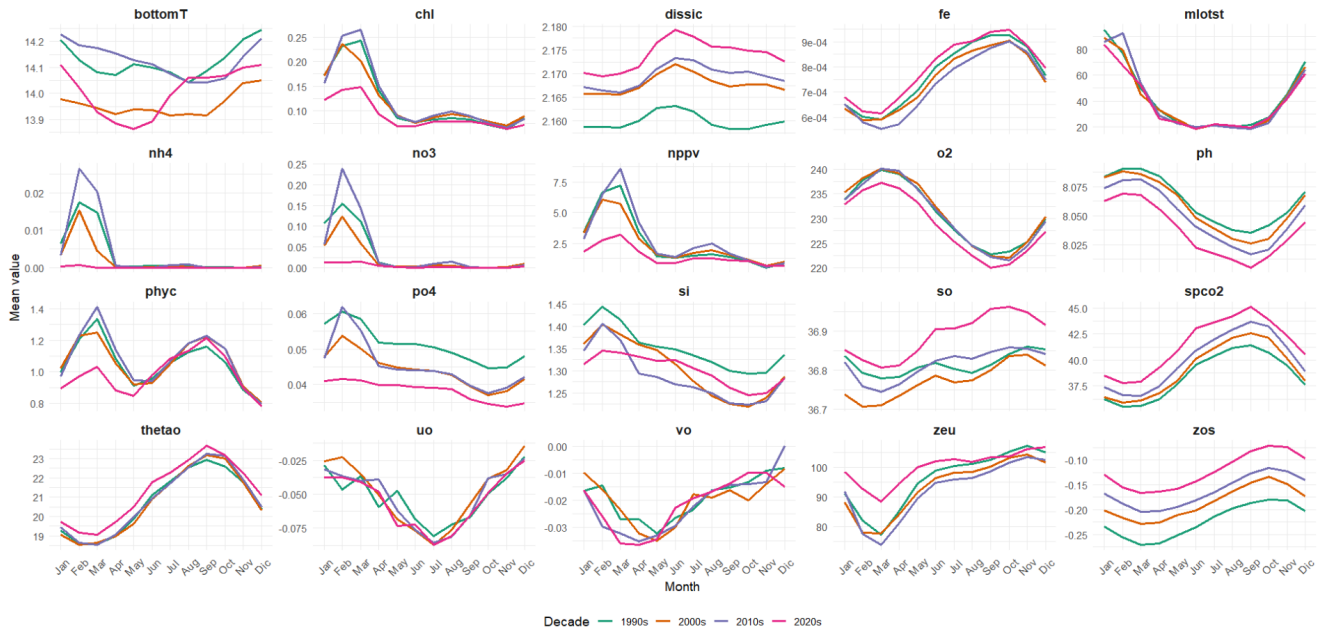


Figure 4. Seasonal and inter-decadal variability of environmental variables: monthly climatologies of selected physical and biogeochemical environmental variables averaged by decade. Lines represent mean monthly values, highlighting seasonal cycles and inter-decadal differences.

NMDS – Diversidad beta entre islas

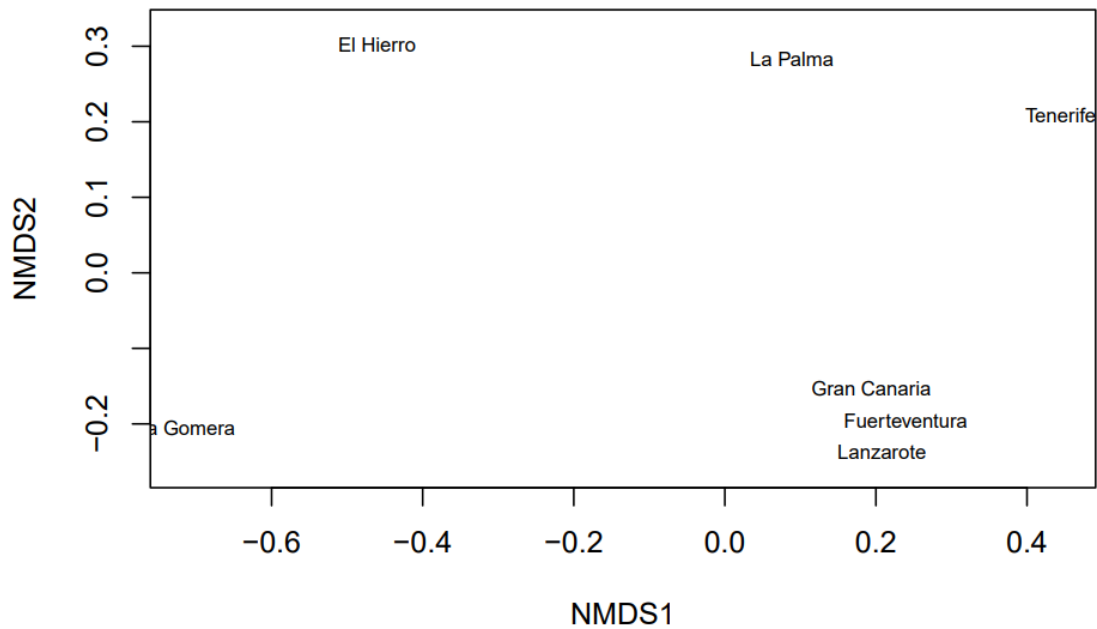


Figure 5. Beta diversity patterns among islands: Non-Metric Multidimensional scaling (NMDS) ordination based on Jaccard dissimilarity of macroalgal species composition among islands. Distances reflect compositional dissimilarity, with clustering indicating greater floristic similarity

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.